

Performance of tracking PV

PVGIS-5 estimates of solar electricity generation

Provided inputs:

Latitude/Longitude: 19.456,72.828
Horizon: Calculated
Database used: PVGIS-ERA5
PV technology: Cryst Sil 2025
PV installed: 1 kWp
System loss: 14 %

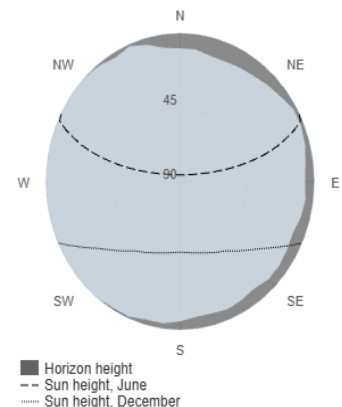
Simulation outputs

Slope angle [°]:
Yearly PV energy production [kWh]:
Yearly in-plane irradiation [kWh/m²]:
Year-to-year variability [kWh]:
Changes in output due to:
Angle of incidence [%]:
Spectral effects [%]:
Temp. and low irradiance [%]:
Total loss [%]:

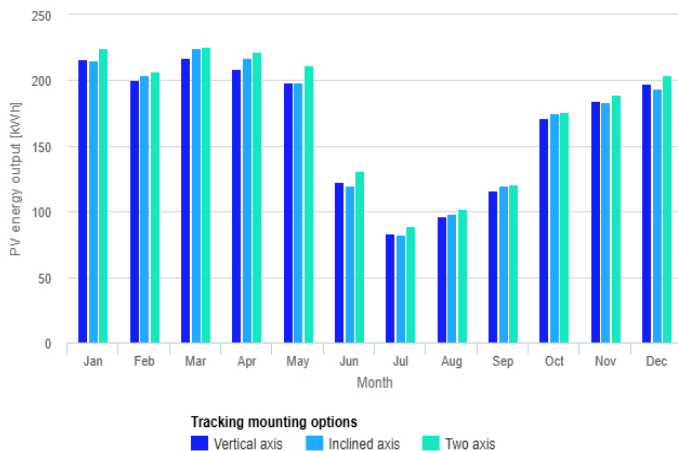
VA*	IA*	2A*
48 (opt)	25 (opt)	-
2012.79	2032.81	2101.71
2572.47	2596.63	2687.97
53.8	54.5	56.9
-1.34	-1.3	-1.25
0.09	0.08	0.09
-7.87	-7.84	-8.01
-21.76	-21.71	-21.81

* VA: Vertical axis
IA: Inclined axis
2A: Two axis

Outline of horizon at chosen location:



Monthly energy output from tracking PV system:



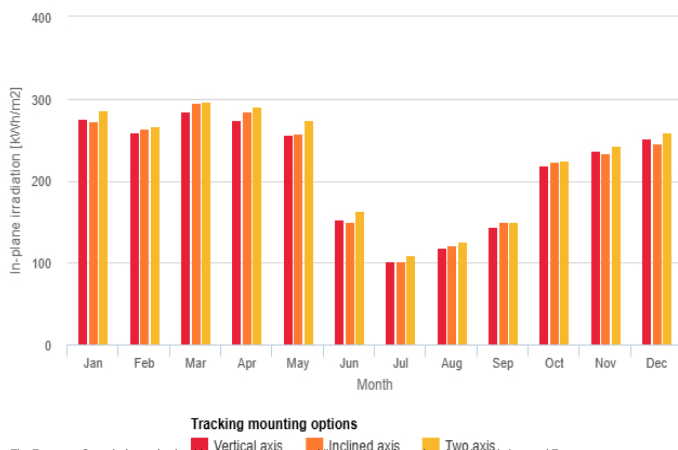
Month	Vertical axis			Inclined axis			Two axis		
	E_m	H(i)_m	SD_m	E_m	H(i)_m	SD_m	E_m	H(i)_m	SD_m
January	216.7	275.3	6.5	215.4	272.8	6.5	224.7	285.4	6.9
February	200.6	258.4	5.6	204.3	263.0	5.8	207.0	266.8	5.9
March	217.1	284.9	7.0	225.1	295.8	7.4	225.4	296.3	7.4
April	209.0	273.5	3.9	216.9	284.2	4.1	221.8	290.9	4.2
May	198.1	256.3	9.3	198.9	257.2	9.3	212.0	274.9	10.1
June	122.5	152.3	15.7	119.8	149.1	15.2	130.7	162.6	16.8
July	83.0	102.1	15.9	82.5	101.5	15.5	88.6	109.0	16.9
August	96.3	118.8	14.6	98.4	121.4	14.8	102.1	126.0	15.4
September	115.7	144.2	16.7	119.8	149.3	17.4	120.7	150.4	17.4
October	171.8	218.5	14.4	175.3	222.9	14.9	176.2	224.2	15.0
November	184.3	236.4	12.1	183.1	234.4	12.1	188.8	242.2	12.7
December	197.8	251.6	9.7	193.4	245.2	9.5	203.8	259.3	10.2

E_m: Average monthly electricity production from the defined system [kWh].

H(i)_m: Average monthly sum of global irradiation per square meter received by the modules of the given system [kWh/m²].

SD_m: Standard deviation of the monthly electricity production due to year-to-year variation [kWh].

Monthly in-plane irradiation for tracking PV system:



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